

AJ PRELIMS 2008 H2 P3

1 Analysis of restriction fragment length polymorphism (RFLP) fragments is an important tool in molecular biology.

(a) (i) What are RFLP fragments?

- DNA fragments from **2 different** genomes/individuals;
- that result from cleavage by **same** restriction enzyme;

(ii) Why do two individuals express different patterns of RFLP fragments?

- 2 organisms/individuals have **diff. VNTR**;
- **differ in distance between restriction sites** of a particular restriction enzyme / number of restriction sites;
- the length / number of the fragments produced **will differ** when the 2 DNA are digested with the **same** restriction enzyme;

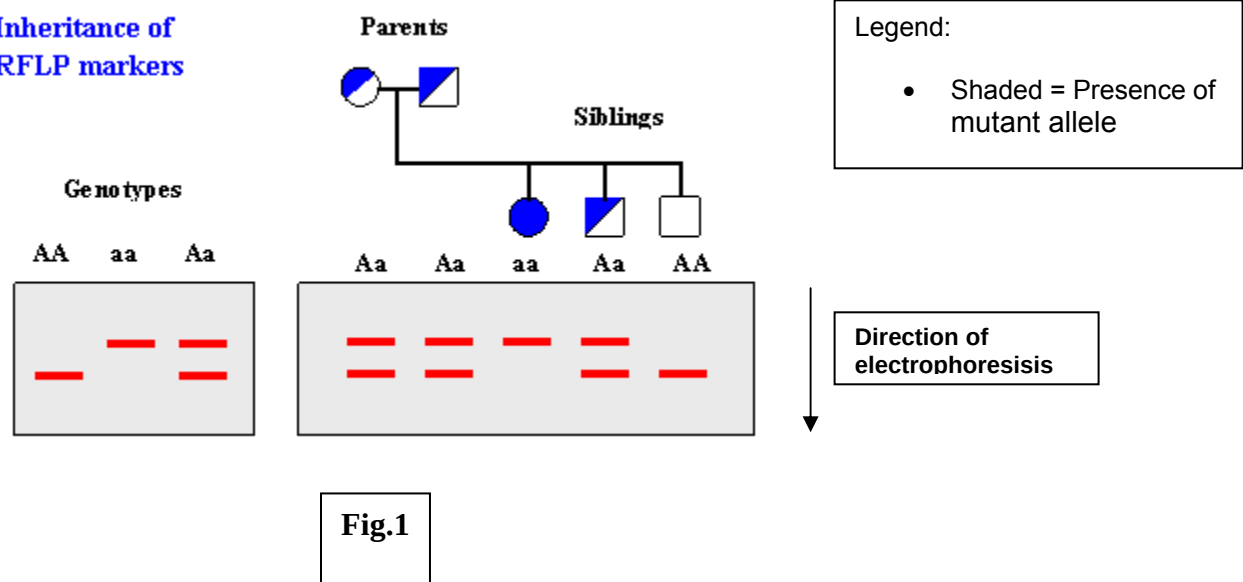
(iii) State two uses of RFLP fragments to society.

- Genomic mapping / diseases detection / DNA fingerprinting etc;

(b) RFLP analysis is used within Southern blotting to check for the presence of a disease-causing mutant allele in a DNA sample.

Fig 1.1 shows the results of Southern blotting for the gene of interest.

Inheritance of RFLP markers



(i) Outline the process of Southern blotting. [5M]

- Restriction endonucleases cut sample DNA at restriction sites;
- The DNA fragments are then electrophoresed on gel to separate them by size;
- (After electrophoresis) The fragments are transferred to a nitrocellulose or nylon membrane;
- The membrane is probed with a **hybridization probe**;
- Pattern of hybridization is visualized by autoradiography / chromogenic detection [1M each]

(ii) What is the mode of inheritance of this disease? [1]

- Autosomal recessive;

(iii) Explain for the bands found in the individuals. [4]

- Alleles 'A' and 'a' differ in DNA sequences / restriction sites
- Allele 'A' produces small DNA fragments when cut with particular restriction enzyme;
- Allele 'a' produces bigger DNA fragments when cut with same restriction enzyme;
- Heterozygote individuals thus express both big and small DNA fragments;

[Total: 15]

- 2 (a) Cystic fibrosis is a genetic disorder caused by a recessive mutation. It is characterized by malfunction of pancreas and abnormal secretion from the lungs and is fatal if symptoms are untreated.

However, heterozygotes for the most common mutant allele causing cystic fibrosis (CF) may be more resistant to bacterial infections of the gut than homozygotes for the normal allele.

(i) Explain how such resistance could affect the frequency of the **mutant** allele in a population. [2]

- Mutant allele confers selective advantage to the individual against bacterial infections of the gut;
- Higher frequency of the heterozygote condition in population (than homozygotes for the normal allele or mutant allele);
- Lower frequency of homozygote mutant alleles because they cause CF and are selected against;

[Any 2]

The bacterium *Salmonella typhi*, which causes typhoid fever in humans, will infect mouse gut cells. Mouse gut epithelium cells were engineered to express either the normal allele or the mutant allele of the gene responsible for cystic

fibrosis. Three strains of *S. typhi* were incubated with cells expressing the mutant CF allele and cells expressing the normal allele. Fig. 2 shows the numbers of bacteria taken up by these cells.

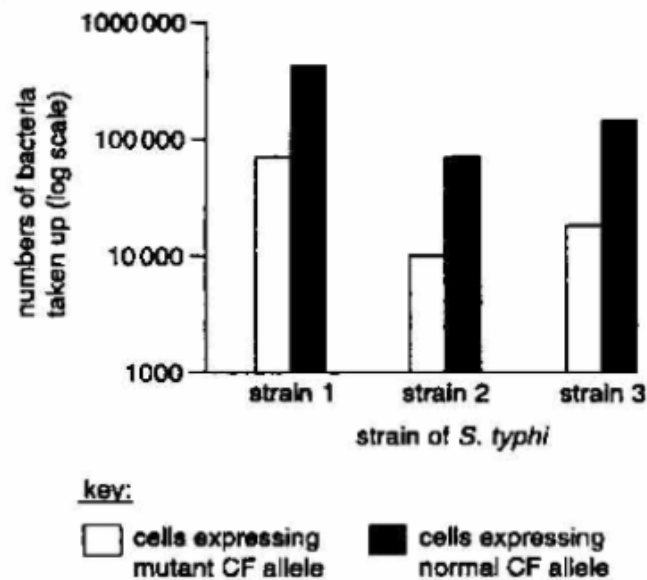


Fig.2.1

(ii) With reference to Fig.2.1, compare the effects of the normal and mutant alleles for cystic fibrosis on the number of bacteria taken up by the mouse gut cells.

- No. of bacteria taken up were less in gut cells expressing mutant CF allele than gut cells expressing normal CF allele for all 3 strains of *S. typhi*;
- Uptake of the bacteria by gut cells is highest in strain 1, and lowest in strain 2 [0.5M]; by ~ 1.8 (if comparing cells expressing mutant alleles) / 9.0 times (if comparing cells expressing normal alleles);
- Bacterial uptake by cells expressing mutant allele is reduced but not prevented;

[Any 3]

(iii) Suggest why the expression of the cystic fibrosis gene affects the number of bacteria taken up.

- Normal CF allele codes for functional cystic fibrosis transmembrane protein on cell surface membrane;
- This allows *S. typhi* to pass through channel;
- Mutant CF allele codes for non functional CFTR protein;

[Any 3]

- (b) Thalassaemia is an inherited autosomal recessive blood disease. In β -thalassaemia, the genetic defect results in reduced rate of synthesis of the β globin chains that make up hemoglobin.

Many cases of the genetic disease β -thalassaemia result from a large deletion in the gene for β globin.

- (i) Explain how electrophoresis allows an allele with a large deletion to be distinguished from the normal allele. [2]

- An allele with a large deletion will have shorter genetic sequences / lighter / fewer fragments;
- DNA fragments will migrate further/faster in gel;

A potential gene therapy for β -thalassaemia involves adding the normal, dominant allele for β globin to the patient's bone marrow cells.

- (ii) Explain why it is theoretically easier to perform gene therapy when a mutant allele is recessive than when it is dominant. [3]

- When mutant allele is recessive, addition of functional/normal dominant allele will mask effect of recessive mutant;
- If mutant allele is dominant, it has to be removed/repaired/inactivated;
- block production of defective gene product

Such gene therapy has been tested in mice. Bone marrow cells carrying the normal allele for human β globin were implanted both in normal mice and in mice heterozygous for the deletion causing β -thalassaemia. Twenty-four weeks later, the percentage of haemoglobin containing two human β globin chains was measured in both types of mouse. The results are shown in Table 2.1.

Type of mouse	Percentage of mouse haemoglobin with two human β globin chains
Normal	13
Heterozygous for the deletion causing β -thalassaemia	24

Table 2.1

(iii) With reference to Table 2.1, suggest why the human gene is expressed less in normal mice.

- Normal mouse contains two β globin alleles with promoters;
- Cells are more likely to transcribe these alleles than the human alleles;
- When mouse is heterozygous / contains one defective β globin allele, the human allele is more likely to be turned on -accounting for 24% of Hb with human globin chains

[Any 2]

[Total: 15]

3 (a) Transgenic plants are created from genetic engineering to introduce desirable traits for economically important plants.

(i) Define transgenic plants and state an example of one economically important transgenic plant. [2]

- Transgenic plants are plants that have been genetically modified by **inserting genes** [0.5M] from **another species / breed** [0.5M]; R: Foreign gene
- Basta-resistant crops / Golden rice / maize with Bt genes / Flavr Savr tomato / Tobacco seeds with phaseolin genes etc / PAT genes in crops for herbicide resistance / Transgenic orchids with firefly luciferase gene;

(ii) Give an example of a desirable trait within the context of an economically important plant. [1]

- Basta-resistant crops – Resistant towards herbicide Basta;
- Golden rice – High in Vit A;
- maize with Bt genes – confers plant resistance towards insects;
- Flavr Savr tomato – higher resistance to spoiling / Picking the fruit while unripe allows for easier handling and extended shelf-life; Tobacco seeds with phaseolin genes – increased nutritional value etc;

[Any 1]

(iii) Outline how the gene that codes for this desirable trait could have been isolated. [4]

- Choose a **restriction enzyme** [0.5M] to excise out **gene of interest and cut gene vector** [0.5M] at restriction sites;
- **Linker DNA** added to **blunt ends** of DNA;
- Add **DNA ligase** to catalyse formation of **phosphodiester bonds** to form recombinant DNA;
- **Insert recombinant DNA** into host cell / **Facilitate transformation** into host cell;

cDNA

-
- Creation of **cDNA** fr mRNA;
 - Use **reverse transcriptase**;
 - Degradation of mRNA;
 - Use of DNA polymerase;

Southern Blotting

- **Restriction enzymes** cut sample DNA at **specific restriction sites**;
- **Electrophoresis** on gel to separate them by **size**;
- Fragments are transferred to a **nitrocellulose / nylon membrane**;
- The membrane is probed with a **hybridization probe**;
- Pattern of hybridization is visualized by autoradiography / chromogenic detection / describe detection method;

PCR

- Use of primers;
 - Use of DNA polymerase;
 - Mention denaturation / annealing / extension
 - Amplify gene of interest
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- Confirm gene by sequencing / phenotypic ID

[Any 4]

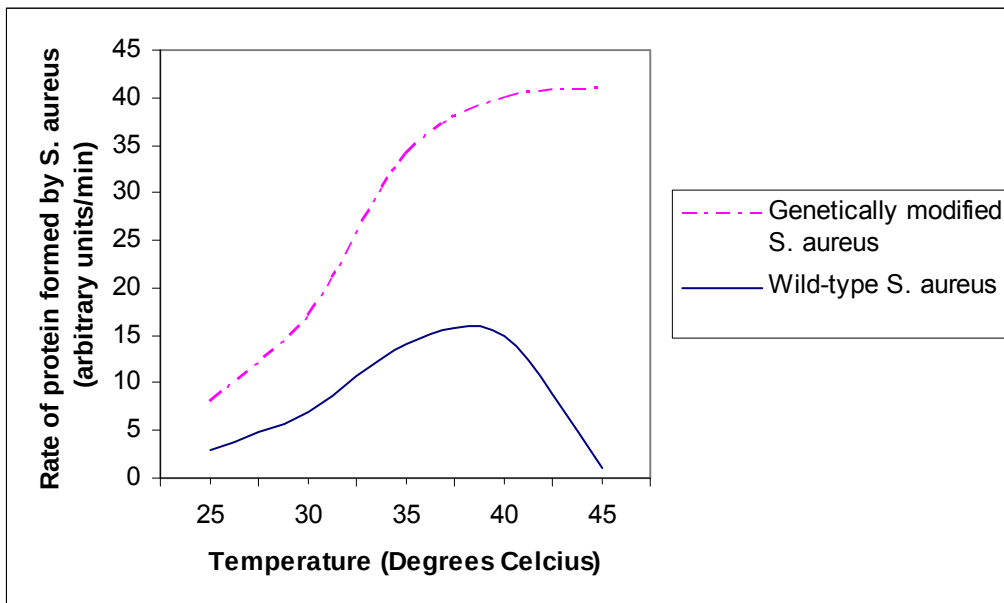
(iv) Explain the use of Ti plasmid in genetic engineering of plants

[3]

-
- Serves as **vector** for **gene of interest**;
 - Gene of interest inserted into **T-DNA region** of plasmid;
 - The resulting **recombinant plasmid** is then placed back into the ***A. tumefaciens*** cell;

(b) *Staphylococcus aureus*, a gram positive bacterium, synthesizes a large number of extracellular proteins that have been postulated to play a role in bacterial virulence. Researchers discovered that the thermostability of some of these proteins differ between the wild type and genetically mutated *S. aureus*. [2]

The following diagram shows the rate of protein production by *S. aureus* at various temperatures.



(i) Describe the effects of increasing temperature on the quantity of product formed for the 2 types of bacteria. [3]

- For wild type, rate of protein formation **increases with temperature** and optimum rate is ~ **16 units/min at ~ 32 °C**;
- For wild type, rate of protein formation **decreases** [0.5M] **beyond optimum temperature** [0.5M];
- For genetically mutated *S. aureus*, rate of protein formation **increases with temperature**;

(ii) Suggest what genetic modifications have occurred to account for the observations in the genetically modified *S. aureus*.

- (Artificially-induced) gene **mutation** in *S. aureus* DNA;
- Codes for **amino acids that form bonds / disulphide bridges** to account for high thermostability in protein; [mention of “bonds” only 0.5M]
- Codes for **efficient protein synthesis machinery** to account for faster protein formation;
- Allows bacteria to be more **resistant to heat** / more active at higher temperatures;

[Any 2] [2]

4	(a)	Describe the characteristics of hematopoietic stem cells.	[8]
		• They are multipotent stem cells; • Organ- or tissue-specific stem cells; • Can produce only cells of a closely related family of cells; • Can differentiate into red blood cells, white blood cells, platelets, etc.; • But cannot develop into other types of cells; • Capable of dividing and; • renewing themselves / have telomerase activity; • They are unspecialized; • Can be found on bone marrow / Peripheral Blood (circulating pool of blood, and not sequestered within the lymphatic system, spleen, liver, or bone marrow) / Umbilical Cord Blood / placental; [Any 8]	
	(b)	Outline how medically important proteins can be mass produced via genetic engineering.	[8]
		• mRNA of protein converted to cDNA by reverse transcriptase; • isolate vector from bacteria; • gene of interest and vector cut with <u>same</u> restriction enzyme to generate complementary sticky ends; • Add DNA ligase to seal the nicks; • form recombinant DNA; • Transform the recombinant DNA into bacteria by electroporation / heat-shock; • Ref. to screening by blue-white selection and replica plating • Scaling-up production of the transgenic bacteria in bio-reactor / fermenter; • Mention of any medically important proteins e.g. HGH, insulin etc; [Any 8]	
	(c)	Briefly discuss two societal concerns arising from the Human Genome Project. Acquisition and use of genetic information • Whether parents have right to have their minor children tested for adult-onset diseases • Argument over how rules should be set for the acquisition and use of genetic information Making decisions • Problem of how to prepare the public to make informed choices • How do we as a society balance current scientific limitations and social risk with long-term benefits Reliability of technology • Query over reliability of genetic tests and • its interpretation by the medical community • Problem of disseminating unforeseen consequences across generations Future generations may feel limited by choices made regarding their genetic traits. • Whether future generations would share their predecessors' views about the desirability of the traits that have been bequeathed to them via GE	[4]

		• Current social-historical trends and biases could place genetic limitations on future generations [Any 4]	
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[Total: 20]